B. Tech. (Computer Engg.) V Semester Examinations

Session 2025-26

Time: 1 hr. **Computer Networks (CSC - 503) Sessional-II** Max. Marks: 12

Note: Attempt All Questions

1. **CO3**	Consider the Cyclic Redundancy Check (CRC) based error detecting scheme having the generator polynomial x^3+x+1. Suppose the message 11000 is to be transmitted. Check bits are appended at the end of the message by the transmitter using the above CRC scheme. Calculate the transmitted bit sequence. Also, show how the receiver will come to know that the error has occurred in the sequence.	**4**	**BL3**
2. **CO3**	A 1 Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 Km and speed of the signal is 3*10^8 m/sec. What should be the packet size for a channel utilization of 25% for a satellite link using Go-Back-127-sliding window protocol?	**4**	**BL5**
3. **CO4**	Define CSMA/CD. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?	**4**	**BL1,4**